AthletixAI

AI-powered sports talent assessment

using nothing but a phone camera

35	33	5	31
SPORTS	KEYPOINTS	METRICS	ENDPOINTS

Archita Tripathi · Manthan Dhanraj

athletixai.onrender.com · live in production

01 — OVERVIEW

Executive Summary

AthletixAI turns any smartphone camera into a sports performance lab. An athlete records a training clip; on-device AI extracts 33 body keypoints, a biomechanics engine measures how they move, and a sport-specific scorer returns a performance report across five metrics — Speed, Agility, Strength, Stamina and Technique — for 35 sports.

The product is live in production and free for athletes. It is built around one principle that shapes every design decision: **the system measures honestly, or it does not measure at all.** The video never leaves the device, scores are deterministic, and the engine refuses to produce a result when it cannot clearly see the athlete.

Attribute	Detail
Category	Vertical AI · sports technology
Status	Live in production, publicly accessible
Users	Athletes, coaches, platform owner
Built by	Archita Tripathi · Manthan Dhanraj

02 — THE PROBLEM

India Doesn't Lack Talent. It Lacks Measurement.

- **No coach** — a rural athlete has no trained eye to tell them what to fix.
- **No baseline** — without numbers, nobody knows if they are improving or plateauing.
- **No visibility** — selectors can only pick from athletes they have physically seen.

A professional biomechanics lab costs lakhs and exists in only a handful of cities, so most athletes are never measured at all. AthletixAI was built so that fewer athletes lose their chance for lack of resources — the founder played competitive cricket and never received the coaching he believed could have taken him further.

03 — THE SOLUTION

A Performance Lab That Fits in a Pocket

An athlete records a drill on any phone and receives a report in seconds, scored across five metrics tuned to their sport:

Metric	What it measures
Speed	Peak movement burst and step cadence
Agility	Body symmetry and direction-change frequency
Strength	Joint flexion depth and range of motion
Stamina	Sustained output across the session

Metric	What it measures
Technique	Movement control, form quality, tracking confidence

Each report also includes sport-specific insights, corrective drills for the weakest metric, injury-watch zones, and a progress trend versus previous sessions.

04 — THE AI

How the AI Actually Works

This is a genuine computer-vision pipeline, not a wrapper around a chatbot. It runs in three stages.

Stage 1 — Pose estimation (real machine learning)

Google's MediaPipe Pose Landmarker, a trained deep-learning model, runs in the browser and returns 33 body landmarks per frame — shoulders, elbows, wrists, hips, knees, ankles — each with an x, y and a visibility confidence.

Stage 2 — Feature extraction (real measurement)

Joint angles are computed with the dot-product formula **cos(θ) = (v1 · v2) / (|v1| × |v2|)** — for example the knee angle from the hip → knee → ankle triangle. The engine also derives frame-to-frame velocity, left/right symmetry, step cadence and range of motion. Frames where a landmark's visibility falls below 0.3 are discarded, so a partly hidden body never pollutes the result.

Stage 3 — Category-aware scoring

Different sports reward different qualities, so a single formula would be wrong: an archer holding perfectly still is performing well; a sprinter standing still is not. The engine carries six scoring profiles, and an effort gate scales the dynamic metrics so low-effort movement cannot reach elite numbers.

Profile	Rewards	Example sports
Precision	Stability, repeatability	Archery, Shooting, Golf
Endurance	Sustained output, cadence	Athletics, Cycling
Combat	Explosive bursts, reaction	Boxing, Wrestling, MMA
Strength	Squat depth, joint range	Weightlifting
Artistic	Flow, balance, control	Gymnastics, Parkour
Dynamic	Balanced profile	Football, Cricket, Tennis

Honesty by design — the system refuses to guess

A tool that invents a number when it cannot see the athlete is worse than no tool at all. The engine returns a clear error instead of a score when there are fewer than 12 frames with a detected person, when the person is visible in under 40% of frames, when there is too little movement data, or when the footage is too blurry (the score is capped, not inflated). There is no random number anywhere in the scoring path — the same clip always produces the same score.

05 — COVERAGE

35 Sports, Across 8 Categories

Category	#	Sports
Team	9	Football, Basketball, Volleyball, Rugby, Hockey, Handball, Lacrosse, Sepak Takraw, Kabaddi
Combat	7	Boxing, Wrestling, MMA, Judo, Karate, Taekwondo, Fencing
Bat & Ball	3	Cricket, Baseball, Softball
Racquet	3	Tennis, Badminton, Table Tennis
Endurance	2	Athletics, Cycling
Precision	5	Archery, Shooting, Golf, Snooker/Pool, Billiards
Artistic	5	Gymnastics, Parkour, Climbing, Skateboarding, Breakdancing
Strength	1	Weightlifting

13 sports were deliberately removed — swimming, diving, equestrian, chess and others — because a phone camera cannot honestly analyse them. Shipping unreliable scores for those would undermine trust in the 35 that work. This was an integrity decision, not a gap.

06 — THE PRODUCT

Features by Role

Athlete

Live pose analysis with skeleton overlay · video upload & analysis · per-sport reports · progress tracking with AI verdicts · career roadmap & weekly plan · shareable athlete card & badges · coach directory, ratings & messaging · leaderboard · PDF export.

Coach

Squad dashboard · full athlete report history · Recruiter Hub with AI Scout picks · head-to-head radar comparison · clickable analytics drill-downs · public coach profile · direct messaging · talent search across the platform.

Owner

Platform overview with 12 live statistics · 14-day activity chart · complete athlete & coach tables · unified activity log. The owner account is hidden from every public list, access is decided entirely server-side, and credentials live only in environment variables — never in source.

07 — ENGINEERING

System Architecture

The system has three layers, and the browser never touches the database directly — it only calls the API, which is what keeps the data layer safe.

Layer	Component	Responsibility
Browser	index.html	Single-file UI, 159 JS functions; MediaPipe runs here
API	app.py (Flask)	31 REST endpoints; auth, CSRF, rate limiting
Data	athletixai.db (SQLite)	12 tables; cascading deletes; indexed reports

Design choice — a single-file frontend

There is no build step and one HTTP request loads the whole app, which matters on slow rural connections. The accepted trade-off: the file is large, so if the team or scope grows a component framework becomes the right call — at today's scale it would be over-engineering.

Data model — 12 tables

users, athlete_profiles, coach_profiles, reports, ai_results, messages, ratings, notifications, email_tokens, activity_logs, plus videos and profile_settings reserved for upcoming features. Key decisions: cascading deletes (removing a user removes all their data — a privacy requirement), a database-level role constraint, case-insensitive email handling, and reports indexed by athlete and date.

08 — SECURITY & PRIVACY

Security and Privacy

Threat	Defence
SQL injection	All queries parameterised — user input never becomes SQL
Cross-site scripting	All user text escaped before it reaches the DOM
CSRF	Token on every state-changing request, matched to the session
Password theft	bcrypt, 12 rounds, unique salt per user
Session hijacking	HttpOnly + SameSite cookies; Secure in production
Broken access control	Role decorators on every route; identity from session, never URL

Privacy by design

The training video never leaves the device — pose analysis runs entirely in the browser and only the resulting numbers are transmitted. This was non-negotiable because most users are minors. A useful side effect: because inference happens on-device, the platform needs no server GPU, so 10,000 athletes cost almost the same to serve as 10. Users can toggle research consent, export all their data as JSON, and

delete their account permanently; under-18 users require guardian consent under the DPDP Act.

09 — STACK

Technology Stack & Scale

Layer	Choice	Why
Backend	Flask (Python)	Small, readable, quick to iterate
Database	SQLite	Zero config; clear path to PostgreSQL
Frontend	Vanilla JavaScript	No build step; one request loads the app
Computer vision	MediaPipe Pose	On-device inference, 33 landmarks
Charts / PDF	Chart.js, jsPDF	Reports, analytics, exports
Hosting	Render + gunicorn	Auto-deploy on push, free TLS

Languages: JavaScript · CSS · Python · HTML · SQL. **Scale:** ~1,300 backend lines · ~3,800 frontend lines · 31 endpoints · 12 tables · 159 JS functions · 35 sports · 33 keypoints. Installable as a Progressive Web App with an offline shell, and a three-language interface (English / Hindi / Hinglish).

10 — HONEST LIMITATIONS

What It Does Not Do Yet

Stated plainly, because a measurement tool that hides its limits cannot be trusted.

- **Rule-based, not trained.** The pose estimation is real ML; the scoring formulas encode domain knowledge, not learned weights.
- **Not yet calibrated.** Proper calibration needs 100+ athletes' clips paired with measured performance.
- **Relative, not absolute.** "Speed improved 8% this month" is dependable; "speed is exactly 82" is an estimate.
- **2D pose has limits.** Depth is unreliable and camera angle matters, so the engine leans on symmetry and relative change.
- **SQLite is single-writer.** Fine for hundreds of concurrent users; PostgreSQL is the next step.
- **Movement, not tactics.** The engine reads movement quality; technique judgement still needs a coach.

11 — ROADMAP

What's Next

Phase	Focus
Now	Cricket bowling module — arm angle vs the ICC 15° legality rule, run-up speed, front-foot landing, follow-through.
Next	Calibration with real academy data; fit scoring weights to measured performance.
Then	Per-sport models, one sport at a time — never claiming accuracy before the data supports it.

Phase	Focus
Later	Academy dashboards, native mobile app, more regional languages.

Talent should not depend on where you were born or who happened to be watching. AthletixAI is free for athletes, always.

Built by Archita Tripathi and Manthan Dhanraj. Built with AI assistance — the architecture, product decisions, debugging and deployment are ours.